Donggyu Joo

Senior Machine Learning Scientist | Samsung Research

Seoul, South Korea | jdg10126@gmail.com

Research interests

Computer vision, vision-language models, image-text retrieval, model compression, knowledge distillation, generative modeling, and on-device AI.

Experience

Samsung Research, Samsung Electronics | Seoul, South Korea
Senior Machine Learning Scientist | September 2022 - Present

• Developed and shipped eight on-device AI models, including Gallery Search and unintended touch detection, deployed across millions to tens of millions of Galaxy devices with my name credited as their developer.

• Develop on-device visual recognition and vision-language models for semantic photo search in Galaxy Gallery.

• Build multimodal data curation, labeling, filtering, training, and evaluation workflows.

• Optimize and deploy models on Galaxy devices, including TFLite conversion, latency tuning, on-device validation, and debugging edge cases from user feedback.

• Develop sequential classification models for unintended pocket-touch detection with real-time mobile CPU inference.

• Lead technical work across data, modeling, evaluation, and release in collaboration with product and engineering teams.

Education

KAIST - Ph.D. in Electrical Engineering | March 2018 - August 2022
Thesis: Effective Deep Neural Network Compression Based on the Network Pruning
Advisor: Prof. Junmo Kim

KAIST - M.S. in Electrical Engineering | September 2016 - February 2018
Thesis: Unconstrained Control of Feature Map Size Using Non-integer Strided Sampling
Advisor: Prof. Junmo Kim

KAIST - B.S. in Electrical Engineering | February 2013 - August 2016
Summa Cum Laude

High School
Korea Science Academy of KAIST (KSA) | 2010 - 2013

Skills

Python, PyTorch, TensorFlow, TFLite, Git; model optimization, quantization, distillation, data curation, robustness evaluation, and on-device deployment.

Publications

* Equal contribution. ICPR 2020 was held in January 2021. Preprints are listed once.

Training Cartoonization Network without Cartoon
Doyeon Kim, Dongyeun Lee, **Donggyu Joo**, Junmo Kim
ICIP 2023 · Paper

Which Metrics for Network Pruning: Final Accuracy? Or Accuracy Drop?
Donggyu Joo, Sunghyun Baek, Junmo Kim
ICIP 2022 · Paper

Linear Combination Approximation of Feature for Channel Pruning
Donggyu Joo, Doyeon Kim, Eojindl Yi, Junmo Kim
CVPR Workshops 2022 · ECV · Paper

Global-Local Path Networks for Monocular Depth Estimation with Vertical CutDepth
Doyeon Kim, Woonghyun Ka, Pyungwhan Ahn, **Donggyu Joo**, Sehwan Chun, Junmo Kim
arXiv preprint · 2022 · Paper

Linearly Replaceable Filters for Deep Network Channel Pruning
Donggyu Joo, Eojindl Yi, Sunghyun Baek, Junmo Kim
AAAI 2021 · Paper

Slimming ResNet by Slimming Shortcut
Donggyu Joo, Doyeon Kim, Junmo Kim
ICPR 2020 · held in 2021 · Paper

Delivering Meaningful Representation for Monocular Depth Estimation
Doyeon Kim, **Donggyu Joo**, Junmo Kim
ICPR 2020 · held in 2021 · Paper

Continual Learning with Extended Kronecker-factored Approximate Curvature
Janghyeon Lee, Hyeong Gwon Hong, **Donggyu Joo**, Junmo Kim
CVPR 2020 · Paper

Residual Continual Learning
Janghyeon Lee, **Donggyu Joo**, Hyeong Gwon Hong, Junmo Kim
AAAI 2020 · Paper

TiVGAN: Text to Image to Video Generation with Step-by-Step Evolutionary Generator
Doyeon Kim*, **Donggyu Joo***, Junmo Kim
IEEE Access · 2020 · Paper

Generating a Fusion Image: One's Identity and Another's Shape
Donggyu Joo*, Doyeon Kim*, Junmo Kim
CVPR 2018 · Paper

Unconstrained Control of Feature Map Size Using Non-integer Strided Sampling
Donggyu Joo, Junho Yim, Junmo Kim
BMVC 2018 · Paper

A Gift from Knowledge Distillation: Fast Optimization, Network Minimization and Transfer Learning
Junho Yim, **Donggyu Joo**, Jihoon Bae, Junmo Kim
CVPR 2017 · Paper

Patents

Neural Network Learning Method and Device for Recognizing Class
Jun-mo Kim, Byung-ju Kim, Joo-chang Kim, Ye-gang Lee, Min-seok Park, Ju-seung Yun, Jae-young Lee, Dong-gyu Joo
US 11,449,733 B2 | Granted patent
Application: 16/645,344
PCT filing: 4 September 2018; US national-stage date: 6 March 2020; Application publication: 10 September 2020; Grant: 20 September 2022
Public record

Electronic Device for Processing Image and Operating Method Thereof
Donggyu Joo, Dongchan Kim, Jinyoung Hwang
US 2025/0292376 A1 | Published application
Application: 19/018,385
Filed: 13 January 2025; Published: 18 September 2025
Public record

Electronic Device and Operating Method for the Same
Sangwon Lee, Sahngmin Yoo, Donggyu Joo, Jiman Kim, Jinyoung Hwang
KR 10-2026-0029843 A | Published application
Application: 10-2024-0114486
Filed: 26 August 2024; Published: 5 March 2026
Public record

Electronic Device Processing Image and Method of Operating the Same
Donggyu Joo, Dongchan Kim, Jinyoung Hwang
KR 10-2026-0006373 A | Published application
Application: 10-2024-0088511
Filed: 4 July 2024; Published: 13 January 2026
Public record

Electronic Apparatus for Identifying Abnormal Touch and Control Method Thereof
Donggyu Joo, Jinyoung Hwang
KR 10-2025-0114797 A | Published application
Application: 10-2024-0009651
Filed: 22 January 2024; Published: 29 July 2025
Public record

Knowledge Transfer Method Using Deep Neural Network and Apparatus Therefor
Junmo Kim, Junho Yim, Donggyu Joo
KR 10-2061935 B1 | Granted patent
Application: 10-2017-0102269
Filed: 11 August 2017; Application publication: 29 August 2018; Registered: 26 December 2019; Grant publication: 2 January 2020
Public record

Knowledge Sharing Based Knowledge Transfer Method for Improving Quality of Knowledge and Apparatus Therefor
Junmo Kim, Junho Yim, Donggyu Joo, Jihoon Bae, Naesoo Kim, Chulsik Pyo
KR 10-2063376 B1 | Granted patent
Application: 10-2017-0117000
Filed: 13 September 2017; Application publication: 29 August 2018; Registered: 31 December 2019; Grant publication: 7 January 2020
Public record

Incremental Training Based Knowledge Transfer Method for Training Large Deep Neural Networks and Apparatus Therefor
Junmo Kim, Junho Yim, Donggyu Joo, Jihoon Bae, Naesoo Kim, Chulsik Pyo
KR 10-2063377 B1 | Granted patent
Application: 10-2017-0117001
Filed: 13 September 2017; Application publication: 29 August 2018; Registered: 31 December 2019; Grant publication: 7 January 2020
Public record